ASSESSMENT TOOL 1 – HIGH (CODING CHALLENGE)

COURSE CODE : CSA09

COURSE NAME : Programming in Java

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1. Library Book Management using Classes & Encapsulation

```
class Book {
    private int bookId;
    private String title, author;
    private double price;
    private boolean issued;
    Book(int id,String t,String a,double p){
        bookId=id; title=t; author=a; price=p;
    }
    void issueBook(){ if(!issued){ issued=true; System.out.println("Issued"); } }
    void returnBook(){ issued=false; System.out.println("Returned"); }
    void display(){ System.out.println(bookId+" "+title+" "+author+" "+price+" "+issued); }
}
public class Main{
    public static void main(String[] args){
        Book b=new Book(101,"Java","James",500);
        b.display(); b.issueBook(); b.returnBook();
    }
}
```

Sample Input

101

Java

James

500

Sample Output

Book Details:

101 Java James 500.0 false

Issued

Returned

2. Employee Payroll System using Inheritance

```
class Employee{ double calculateSalary(){ return 0; } }
class PermanentEmployee extends Employee{
    double basic=50000;
    double calculateSalary(){ return basic+10000; }
}
class ContractEmployee extends Employee{
    double days=25;
    double calculateSalary(){ return days*1000; }
}
public class Main{
    public static void main(String[] args){
        Employee e1=new PermanentEmployee();
        Employee e2=new ContractEmployee();
        System.out.println(e1.calculateSalary());
        System.out.println(e2.calculateSalary());
    }
}
```

Sample Input

(No Input)

Sample Output

Permanent Salary = 60000.0

Contract Salary = 25000.0

3. Bank Account System using Data Hiding

```
class BankAccount{
    private double balance=1000;
    void deposit(double a){ balance+=a; }
    void withdraw(double a){ if(a<=balance) balance-=a; }
    void show(){ System.out.println(balance); }
}
public class Main{
    public static void main(String[] args){
        BankAccount b=new BankAccount();
        b.deposit(500); b.withdraw(200); b.show();
    }
}
```

Sample Input

Deposit:500

Withdraw:200

Sample Output

Balance = 1300.0

4. Shape Area Calculator using Abstract Class & Polymorphism

```
abstract class Shape{ abstract double area(); }
class Circle extends Shape{
    double r=5;
    double area(){ return 3.14*r*r; }
}
class Rectangle extends Shape{
    double l=4,b=6;
    double area(){ return l*b; }
}
public class Main{
    public static void main(String[] args){
        Shape s1=new Circle();
        Shape s2=new Rectangle();
        System.out.println(s1.area());
```

```
        System.out.println(s2.area());
    }
}
```

Sample Input

(No Input)

Sample Output

Circle Area = 78.5

Rectangle Area = 24.0

5. Student Result Analyzer using Arrays of Objects

```
class Student{
    int roll; String name; int m1,m2,m3;
    Student(int r,String n,int a,int b,int c){
        roll=r; name=n; m1=a; m2=b; m3=c;
    }
    int total(){ return m1+m2+m3; }
}
public class Main{
    public static void main(String[] args){
        Student s=new Student(1,"Sana",90,85,95);
        System.out.println("Total = "+s.total());
    }
}
```

Sample Input

1 90 85 95

Sample Output

Total = 270